<u>HW 1 Instruction Manual</u>

EE 134 Digital IC Design
Winter 2026
University of California, Riverside

<u>Access to LTspice and Transistor Models</u>

For this course, we will primarily use LTspice for circuit-level simulations.

To access the simulator, go to:

https://www.analog.com/en/resources/design-tools-and-calculators/ltspice-simulator.html

Download the most recent version and follow the instructions to install on your computer.

For the transistors, we will be using a commercial library: GlobalFoundries 180nm process. It is also called a PDK (process design kit).

The library is provided as a zip file on Canvas, under xxx. Download it too, unzip it, and keep it on your computer. Inside this folder called gf180_pdk, you will see multiple .txt files, each of which models some devices like transistor or resistor in this PDK. For this course, we will use **cmos_tt.txt** unless otherwise specified. Processes with the suffix of _t stand for typical in the process corner. cmos_tt then means both the NMOS and PMOS are typical. _s and _f respectively standards for slow and fast, so the devices have slightly different characteristics.

<u>Using LTspice</u>

In LTspice, make sure you include the SPICE directive of ".inc xxx\gf180_pdk\cmos_tt.txt", where xxx is the full directory path to your PDK folder on your computer. For each NMOS and PMOS device we use, remember to change the model name to **nmos_3p3** and **pmos_3p3** respectively.

Some good practices to avoid bugs:

1) Don't leave terminals floating.
2) Make sure terminals that should be grounded are indeed grounded.
3) Make sure body terminals are connected somewhere (by default, PMOS to VDD and NMOS to GND)
4) Initialize voltage level in each voltage source, even if you are sweeping it in DC analysis.

Some tutorials can be found in Canvas. Feel free to also search for other resources online for tutorial and debugging.

It may be more convenient to create a separate .asc file for each question that requires LTspice.

Part 1: MOSFET Analysis

Q1.1: The following parameters are given. No simulation is required for this question.

NMOS:
- V_{GS} = 1.8 V
- $V_{th,n}$ = 0.7 V
- $\mu_n C_{ox}$ = 200 µA/V^2
- W/L = 10

PMOS:
- V_{SG} = 1.8 V
- $|V_{th,p}|$ = 0.9 V
- $\mu_p C_{ox}$ = 80 µA/V^2
- W/L = 20

Assume both are operating in the saturation region. Compute the current ID for both devices. Show your work. Explain why PMOS devices often have wider channels in digital cells.

Q1.2: In LTspice, simulate and plot the ID-VG curves for NMOS. Size NMOS to be W=2um and L=1um. VG should be swept from 0V to 3V with 0.01V increment. VD should be fixed at 3V. Show a screenshot of the curve.

Q1.3: In LTspice, simulate and plot the ID-VG curves for PMOS. Size PMOS to be W=4um and L=1um. VG should be swept from 0V to 3V with 0.01V increment. VD should be fixed at 3V. Be careful about the signs of PMOS (i.e., sweep the gate from 3 V down to 0 V). Show a screenshot of the curve.

Q1.4: This question requires the completion of Q1.2/1.3. With your simulated ID-VG curves, estimate the threshold voltages for NMOS and PMOS. There are several ways to estimate the threshold voltage of a device. Here we will use the square root of ID current method. On your waveform, you should see a legend of Id(M1), assuming that your device is named M1. Right-click on it, and you will see a place where you can alter the algebraic expression of this waveform. Type in sqrt(Id(M1)), and your curve will be updated to reflect sqrt(Id) vs. VG. You will also see that there will be a somewhat linear slope right after the knee (the inflection point). With the method of your choice (export this figure to Paint and draw a line; use a ruler on your monitor etc.), extend this line towards the x-axis. The intercept to the x-axis (VG) will be your estimated Vth. Do this for both NMOS and PMOS, again be careful with the sign convention and polarity of PMOS threshold voltage. Explain your method of extrapolation, report the estimated Vth for NMOS and PMOS. Finally, compare your result with the nominal Vth listed in the SPICE models (~0.71V for NMOS and ~|0.92V| for PMOS).

Q1.5: Simulate and plot the ID-VD curves for NMOS. This requires a DC sweep of both VG and VD together. For VG, sweep from 1 to 3V with 0.5V increment. For VD, sweep from 0 to 3V with

0.01V increment. Size NMOS to be W=2um and L=1um. If your simulation is correct, your waveform should contain multiple curves at different VG values. For each curve, identify the transition point from the linear to saturation region. Explain your method of approximating the transitions.

Part B: Inverter Analysis

Q1.6: In LTspice, build a CMOS inverter with NMOS (2um/1um) and PMOS (4um/1um). No need to attach an extra capacitor to the inverter output. Perform DC sweep from 0 to 3V, with increment of 0.01V. Show the waveform with Vin and Vout on the same plot. Report where the input and output voltages are the same. You will likely notice that this intersection is not located at half VDD. Explain why not.

Q1.7: With the same setup as in Q1.6 and NMOS fixed at (2um/1um), tweak the PMOS width (keep length L the same) so that the intersection of input and output voltages is located at half VDD (1.5V). Show the waveform with Vin and Vout on the same plot, and report the updated sizing for the PMOS.

Q1.8: With a CMOS inverter with NMOS (2um/1um) and PMOS (4um/1um), we will run the transient simulation. Keep the same VDD of 3V. You need to change the input voltage supply into a PULSE voltage source: PULSE(0 3 5n 1n 1n 8n 20n). Also add a 50 fF capacitor to the inverter output. Run transient simulation for 20 ns. Show the waveform with both the input and output voltages over time. Next, report the low-to-high delay and high-to-low delay. These are characterized from midpoint to midpoint (for example, 1.5V point of input to 1.5V point of output). You may use the signal cursors to better locate these points.

Q1.9: We will estimate the input capacitance of an inverter using transient simulation. In this method, we will need 3 inverters: A, B, and C. For all inverters, size the same as before: NMOS (2um/1um) and PMOS (4um/1um). Inverter A maintains the same setup as in Q1.6, driving an added capacitance Cload. Inverter B will drive the input of inverter C (called a two-stage inverter chain) and no other load. Inverters A and B can share the same PULSE input supply. All 3 inverters use the same VDD of 3V. With this setup, run the transient simulation, plot the PULSE signal, inverter A's output, and inverter B's output. You may notice that inverters A and B have different delays. Slowly tweak inverter A's Cload until their high-to-low and low-to-high delays match. Show the waveform with the finalized Cload. Also report this Cload value, which is the estimated input capacitance of inverter C.

Q1.10: Repeat the steps of Q1.9, with the only difference of changing inverter C's dimension to NMOS (4um/1um) and PMOS (8um/1um). Do the same estimation for the new inverter C's input capacitance with inverter A's Cload. Show the waveform with the finalized Cload where the delays match, and also report this Cload value. Explain why this new inverter C's input capacitance is different from that in Q1.9.

<u>Submission Instruction</u>

All solutions (for both analytical questions and explanations of simulation results) must be typed. No handwriting. For any equations, you may use the "Equation" feature in Word.

The texts and waveforms in all screenshots of LTspice results must be readable.

Submit the report that documents each question's required items in a PDF file. The report should include your name. The PDF is named hw1_firstname_lastname.pdf. Upload the report to Canvas assignment.

<u>Grading</u>

Each question in this homework is worth 2 points (Q1.1 is worth 2 points, Q1.2 is worth 2 points etc.). Some questions have multiple small parts. Grading for each question is given in a ternary format: 0 (missing or entirely incorrect), 1 (partially correct with major errors or missing justification), or 2 points (correct approach with correct result and clear explanation).